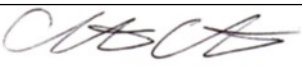


Guideline Title: Defibrillation

Guideline Number: GUID-3203-PR008

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Restoration of a perfusing cardiac rhythm in a patient exhibiting pulseless ventricular tachycardia or ventricular fibrillation
- II. **DEPARTMENT GUIDELINE:**
 - A. **EQUIPMENT:**
 - i. Cardiac monitor/defibrillator. Current department monitor used is the Zoll X Series.
 - ii. Conductive hands-free defibrillation pads
 1. Adult
 2. Pediatric pads for infants/children under 15 kg.
 - B. **INDICATIONS:**
 - i. Ventricular fibrillation
 - ii. Pulseless ventricular tachycardia.
 - C. **CONTRAINDICATIONS:** Patient is wet or for any other reason cannot be electrically isolated from the medical team members.
 - D. **PROCEDURE:**
 - i. If the procedure is to be performed in flight, notify the pilot of the intent to cardiovert. Do NOT proceed further until authorized to do so by the pilot.
 - ii. Place conductive hands-free pads on the patient's bare chest per manufacturers guidelines.
 1. Anterior / Posterior: place negative electrode on left anterior chest, halfway between the xiphoid process and the left nipple. This corresponds with the V2-V3 ECG position. Place the positive electrode on left posterior chest beneath the scapula and lateral to the spine.
 2. Anterior / Anterior: place the negative electrode on the left chest, mid-axilla over the fourth intercostal space. Place positive electrode on anterior right chest, sub-clavicular area.
 3. Avoid placing the pads over any implantable electrical device.

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- iii. Confirm the need to defibrillate/deliver unsynchronized shock per *GUID3203-C007 Ventricular Fibrillation and Pulseless Ventricular Tachycardia*.
- iv. Turn the monitor/defibrillator on. Locate defib mode if using a non-department monitor.
- v. Select the joule dose by adjusting the energy level arrow buttons based on current recommended dosing.
 - 1. Adults: 120-200 J for biphasic
 - 2. Pediatric: Defibrillation at 2 J/kg
- vi. Charge the monitor/defibrillator to the appropriate energy level.
- vii. When defibrillator is fully charged, notify team members of the intent to defibrillate by stating loudly and clearly "CLEAR for shock" and assuring that nobody is touching the patient or stretcher.
- viii. Push the SHOCK button to discharge the shock, defibrillating the patient.
- ix. Immediately after shock delivery, resume CPR beginning with compressions for 5 cycles (about 2 minutes), and then recheck rhythm. Interruption of compressions should be minimized.
- x. Observe for any change in end-tidal CO₂.
- xi. Repeat as appropriate for the patient's condition. Second and subsequent doses should be equivalent or higher.
 - 1. Adult: Up to the highest setting of 200 J for the Zoll X monitor.
 - a. Max setting may be higher for other facility/agency monitors e.g., 360 J.
 - 2. Pediatric: Second shock at 4 J/kg. Subsequent shocks ≥ 4 J/kg to a maximum of 10 J/kg or adult dose.
- xii. After three unsuccessful defibrillation attempts in an adult patient, if a second monitor is available, apply a second set of pads in the anterior-posterior (or anterior-anterior if A/P placement was used first) connected to the second monitor and perform dual sequential defibrillation (DSED) by pushing the shock button on the second monitor 1 second after pressing shock on the initial monitor (use maximum setting, up to 360 J, on both monitors)

III. DOCUMENTATION:

- A. EMS charts

IV. REFERENCES:

- A. American Heart Association, 2015.